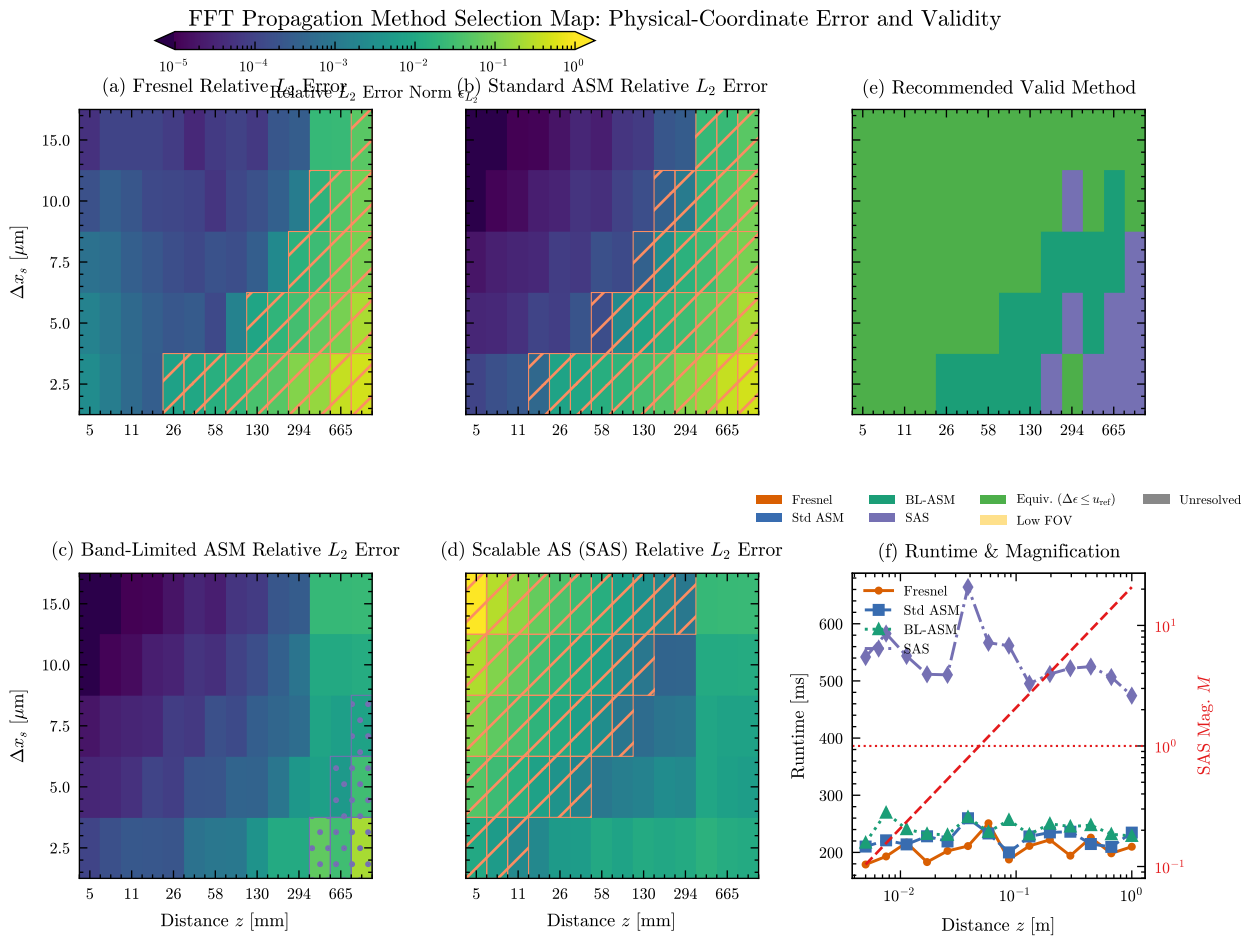

FFT Propagation Method Selection Map: Physical-Coordinate Error and Validity
Relative $L_2$ Error Norm $\epsilon_{L_2}$
$10^{-5}$ $10^{-4}$ $10^{-3}$ $10^{-2}$ $10^{-1}$ $10^{0}$
(a) Fresnel Relative $L_2$ Error
(b) Standard ASM Relative $L_2$ Error
(e) Recommended Valid Method
(c) Band-Limited ASM Relative $L_2$ Error
(d) Scalable AS (SAS) Relative $L_2$ Error
(f) Runtime & Magnification
$\Delta x_s$ [$\mu$m]
15.0
10.0
7.5
5.0
2.5
5 11 26 58 130 294 665
Distance $z$ [mm]
Fresnel
Std ASM
BL-ASM
SAS
Equiv. ($\Delta\epsilon \leq u_{\text{ref}}$)
Low FOV
Unresolved
Runtime [ms]
600
500
400
300
200
$10^{-2}$ $10^{-1}$ $10^{0}$
Distance $z$ [m]
SAS Mag. $M$
$10^{1}$
$10^{0}$
$10^{-1}$